

BePro AI/ML — Olist Compound Curriculum

One dataset, one project, six modules of compounding work, one resume-grade portfolio piece

Saidazam — Mentor & Product Owner

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Olist Compound Curriculum (Modules 3 → 8)

Premise. Every student walks out of the program with **one real GitHub repository** that demonstrates an end-to-end ML system on the Olist Brazilian e-commerce dataset. Each module adds one layer to the same project. By the end, the project is a CV bullet, not a homework folder.

0. Why this design

0.1 What was broken

- Each module had its own toy dataset → no continuity, no compounding skill.
- Cleaning class used a clean dataset (theatre).
- Capstone-style assignments only existed at M8 → too late for portfolio building.
- Students with weak English / first-time laptop users (Type 2) drowned because every module asked them to learn a new domain.
- Strong students (Type 1) felt the work was repetitive because new domain new skill.

0.2 What changes

- **Single dataset, locked from M3 on:** Olist Brazilian E-Commerce (Kaggle).
- **Cumulative work product:** the cleaned dataset from M3 feeds the M4 classifier feeds the M5 segmentation feeds the M6 neural net feeds the M7 NLP layer feeds the M8 deployed API.
- **Two assignment tiers, same dataset:** Bronze (Type 2 — A2 English, beginner laptop) and Gold (Type 1 — comfortable, wants stretch). Same rubric, different depth. Bronze never feels stupid. Gold never feels under-challenged.
- **Every class previews the assignment 1-2 weeks ahead** so students know *why* today matters.

0.3 What stays the same

- The 8-module topic structure is unchanged.
- Module 1 (concepts) and Module 2 (Python tooling) stay as written. Olist enters at M3.
- Class hours, exam dates, lab format unchanged.
- Submission workflow (PR per assignment) unchanged.

1. The dataset — Olist Brazilian E-Commerce

Source: <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce> **License:** CC BY-NC-SA 4.0 (free for educational / non-commercial use) **Size:** ~50 MB, 9 CSV files **Period covered:** 2016-09 → 2018-10

1.1 Files students get

File	Rows	Why we use it
olist_customers_dataset.csv	99,441	Customer ID linking, geography
olist_orders_dataset.csv	99,441	Core order timestamps + status (the spine)
olist_order_items_dataset.csv	112,650	Per-product line items, freight, prices
olist_order_payments_dataset.csv	103,886	Payment method, value, instalments
olist_order_reviews_dataset.csv	99,224	5-star rating + free-text review (Portuguese) — the M7 NLP source
olist_products_dataset.csv	32,951	Category, dimensions, weight
olist_sellers_dataset.csv	3,095	Seller geography
olist_geolocation_dataset.csv	1,000,163	Lat/lon for distance calculations
product_category_name_translation.csv	7	Portuguese → English category names

1.2 Why Olist specifically (over telco, retail, etc.)

- **Genuinely dirty.** Mixed date formats. Missing delivery dates (~3% — these are the *interesting* rows: orders that never arrived). Inconsistent state-code casing. NaN review comments. Real noise that real cleaning skills fix.
- **Has all three data shapes.** Numeric (price, freight), categorical (state, payment_type), text (review_comment_message), datetime (5+ timestamps per order). One dataset → covers M3-M7 needs.
- **Multi-table relational.** Forces students to learn merging properly (M2 skill) before they can model.
- **Multiple targets to choose from** — covered in §2.
- **Distance calculation works.** Brazil is huge, geolocation is real, Haversine over customer × seller cities is a real signal. Perfect for M3 feature engineering.
- **Resume-friendly headline.** “Built an end-to-end ML system on a 100k-order Brazilian e-commerce dataset” beats “predicted churn on a synthetic telco dataset” every time. Hiring managers know Olist.

2. Targets — what we predict / segment / explain

Target	Type	Modules where used
is_late (delivered_customer_date > estimated_delivery_date)	Binary	M4 (classifier), M6 (neural net)

Target	Type	Modules where used
<code>review_score</code> (1-5 stars)	Multi-class	M4 stretch, M6 stretch
<code>customer_segment</code> (clusters of customer behaviour)	Unsupervised	M5
<code>review_sentiment</code> (positive / negative / neutral from review text)	NLP-derived	M7
<code>will_repurchase</code> (<code>customer_unique_id</code> appears > 1 time within 90 days)	Binary	M5 stretch, M8 capstone

The **headline target for the compound project is `is_late`** — chosen because: - Imbalanced (~7% late) → forces M4 students to confront accuracy-as-wrong-metric. - Real business value → “predict which orders will arrive late so support can warn customers.” - Improves with every layer: clean data → classifier → embedded neural net → enriched with review sentiment → deployed.

3. Module-by-module compounding

Raw 9-CSV Olist dump

M3	Clean + merge + feature-engineer One canonical analytical table	→ <code>olist_clean.parquet</code> (the spine output)
M4	Train classifier on <code>is_late</code> Logistic, Trees, Random Forest	→ <code>model_v1.pkl</code> + evaluation report
M5	Customer segmentation K-Means, dimensionality reduction	→ <code>customer_clusters.csv</code> + 4 named cohorts
M6	Neural net classifier Beat or match M4 baseline	→ <code>model_v2_nn.pkl</code> + comparison
M7	Review-text sentiment + add to model TF-IDF / embeddings → enriched data	→ <code>model_v3_with_text.pkl</code>
M8	Deploy as Flask/FastAPI dashboard	→ live URL + GitHub repo + README

3.1 Module 3 — Data Preparation & Feature Engineering

Lecture topic stays: cleaning, encoding, feature engineering, feature selection, pipelines, EDA lab.

What's new: - Every class uses raw Olist tables. - Every assignment writes one piece of `clean_olist.py`. - The Module 3 lab (Class 6) outputs `olist_clean.parquet` — the file every subsequent module loads as input.

Cumulative deliverable at end of M3: a single Parquet file (~50 MB) containing one row per order, with columns: `order_id`, `customer_id`, `customer_state`, `seller_state`, `distance_km`, `freight_value`, `num_items`, `total_price`, `payment_type`, `review_score`, `delivered_days`, `estimated_days`, `is_late`, `purchase_hour`, `purchase_dayofweek`, `purchase_month`.

Type 2 (Bronze) homework version: “Here’s a notebook with all the cleaning code; run it, fix the 5 TODOs, and produce the Parquet.” **Type 1 (Gold) homework version:** “Given the 9 raw CSVs, build the same cleaning + feature pipeline yourself, document each decision.”

3.2 Module 4 — Supervised Learning

Lecture topic stays: linear regression, logistic regression, trees & random forests, evaluation metrics, overfitting, classifier lab.

What's new: - Every example, every assignment loads `olist_clean.parquet` from M3. - Class 4-1 (Linear Regression) → predict `freight_value` from `distance_km` (concrete, intuitive). - Class 4-2 (Logistic Regression) → predict `is_late` (the headline target). - Class 4-3 (Trees / Random Forest) → improve the `is_late` baseline. - Class 4-4 (Evaluation) → confront accuracy on the imbalanced `is_late` (only 7% positives) → introduce precision/recall/F1/ROC-AUC. - Class 4-5 (Overfitting) → cross-validation on the same task; compare M4-3 RF vs cross-validated RF. - Class 4-6 (Lab) → ship `model_v1.pkl` + a 1-page report.

Cumulative deliverable at end of M4: trained model, evaluation report, and a comparison table (Logistic vs Tree vs Random Forest with metrics).

3.3 Module 5 — Unsupervised Learning

Lecture topic stays: clustering intro, K-Means, hierarchical, dimensionality reduction, anomaly detection, segmentation lab.

What's new: - Class 5-1 (clustering intro) → why segment Olist customers? - Class 5-2 (K-Means) → cluster customers by RFM (Recency from `order_purchase_timestamp`, Frequency = order count, Monetary = sum of `total_price`). Output: 4 named cohorts. - Class 5-3 (hierarchical) → repeat with hierarchical, compare dendrograms. - Class 5-4 (PCA / t-SNE) → visualise the cohorts in 2D. - Class 5-5 (anomaly detection) → find unusual orders (very high freight relative to value, etc.). - Class 5-6 (lab) → output `customer_clusters.csv` joining cluster ID back to the M3 Parquet.

Cumulative deliverable: `customer_clusters.csv` + a one-page cohort write-up (“Cohort A: high-value, low-frequency luxury buyers; recommend X”).

3.4 Module 6 — Neural Networks & Deep Learning

Lecture topic stays: perceptrons, backprop, CNNs, RNNs, transfer learning, image classifier lab.

What's new (key adaptation — most NN topics fit Olist tabular data weirdly, so we tier the assignments): - Class 6-1 (perceptrons) → build a 1-layer NN that matches logistic regression on `is_late`. Prove the theoretical equivalence. - Class 6-2 (backprop) → train a 3-layer MLP on `is_late`. Compare to M4 random forest. - Class 6-3 (CNNs) → demo on **product photo** task (Olist product images dataset is available as a separate Kaggle download). Optional for Bronze, required for Gold. - Class 6-4 (RNNs) → demo on **review text sequences** (briefly — the heavier text work is in M7). - Class 6-5 (transfer learning) → fine-tune a small pretrained model on the product images (Gold only). - Class 6-6 (lab) → ship `model_v2_nn.pkl` for tabular `is_late` + a head-to-head comparison with M4 random forest.

Cumulative deliverable: trained NN, comparison report (which model wins, why).

3.5 Module 7 — NLP & Computer Vision

Lecture topic stays: text preprocessing, embeddings, transformers, computer vision, working with LLMs, sentiment/object lab.

What's new: - Class 7-1 (text preprocessing) → clean and tokenise the `review_comment_message` column from Olist (Portuguese). - Class 7-2 (embeddings) → TF-IDF and Word2Vec on the reviews. - Class 7-3 (transformers) → use a pre-trained multilingual sentiment model (e.g. `distilbert-base-multilingual-cased`) to score each review. - Class 7-4 (computer vision) → CV pivot: classify Olist product images by category. (Different sub-dataset, same project narrative.) - Class 7-5 (working with LLMs) → use an LLM API to summarise the top complaints per product category. - Class 7-6 (lab) → output `model_v3_with_text.pkl` — the M4 model enriched with `review_sentiment` as a new feature.

Cumulative deliverable: sentiment-enriched model + lift comparison (does adding text help?).

3.6 Module 8 — Capstone & Deployment

Lecture topic stays: project planning, model dev sprint, Flask/FastAPI deployment, MLOps, demo prep, final presentations.

What's new: - Class 8-1 (planning) → write a 1-page architecture doc for the deployed system (input → model → output). - Class 8-2 (model dev sprint) → consolidate M4-M7 outputs into one polished model. - Class 8-3 (deployment) → wrap the model as a Flask/FastAPI endpoint. Input: order JSON. Output: `is_late` probability + reason. - Class 8-4 (MLOps) → add basic logging + a `/health` endpoint + a re-train script. - Class 8-5 (demo prep) → record a 3-minute video walkthrough. - Class 8-6 (final presentations) → live demo in front of the cohort.

Final cumulative deliverable: a public GitHub repo + a deployed URL + a 3-minute video.

4. The compound resume bullet

Every student at end of program can write this on their CV (with their own results filled in):

Olist E-commerce Delivery & Reviews Intelligence Platform (BePro AI/ML, 2026) Built an end-to-end ML system on the Olist Brazilian e-commerce dataset (100k+ orders). Engineered a relational analytical table (15 features) from 9 raw CSVs. Trained gradient-boosted classifier predicting late deliveries (ROC-AUC X.XX). Segmented customers into 4 RFM cohorts. Replaced the classifier with a 3-layer neural net for a Y% lift in F1. Enriched the model with review-text sentiment from a multilingual transformer for an additional Z% gain. Deployed the final model as a FastAPI endpoint with logging and re-train pipeline. Stack: Python · Pandas · NumPy · scikit-learn · TensorFlow · Hugging Face · FastAPI · Docker. Repo: github.com/olist-delivery-intelligence

This bullet sells. Hiring managers can read it.

5. Two-tier assignments — Bronze vs Gold

Every assignment from M3 onwards has both versions, **same dataset, same rubric, different depth**.

Aspect	Bronze (Type 2)	Gold (Type 1)
Code skeleton	50% pre-filled with # TODO markers	Empty notebook + spec only
Steps	Numbered, in order	Free choice of order + approach
Hints	Inline comments explaining each line	None — discover via docs
Deliverable	The expected output file	Same output file + a written justification of design choices

Aspect	Bronze (Type 2)	Gold (Type 1)
Time budget	1.5 × the gold version	1×
Grading	Same rubric. Bronze max grade is 90% (10% reserved for extension work students self-select).	Full 100% available.

Student picks their own tier each assignment. Switching tiers between assignments is allowed (encouraged — Bronze students aim for Gold by M5).

6. Implementation timeline (mentor-side)

Week	Milestone	Status
Week 0 (this week)	Master plan written + M3 content rewritten	in progress
Week 1	M3 lab data ready (clean Olist Parquet generation script in repo)	not started
Week 2	M4 content rewritten with Olist-grounded examples	
Week 3	M4 lab handout + Bronze/Gold versions	
Week 4 (mid-term week)	Pause, mid-term covers M1-M4	scheduled
Week 5-6	M5 rewrite (segmentation on the M3 Parquet)	
Week 7-8	M6 rewrite (NN on the M3 Parquet)	
Week 9-10	M7 rewrite (NLP on Olist reviews + CV on product photos)	
Week 11-12	M8 rewrite (deployment of the compound model)	
Week 13	Final exam + demo day	scheduled

7. Repository structure (per cohort, going forward)

```

bepro-aiml/<cohort>/
  olist_data/                                ← single canonical dataset, committed once
    olist_customers_dataset.csv
    olist_orders_dataset.csv
    ...
    README.md (license, source, schema)
  olist_pipeline/                             ← grows module by module
    module_3_clean.py                         ← M3 lab output: cleans 9 CSVs → olist_clean.parquet
    module_4_classifier.py                    ← M4: trains on olist_clean.parquet
    module_5_segments.py                     ← M5: clusters customers
    module_6_nn.py                           ← M6: replaces classifier with NN
    module_7_text.py                          ← M7: adds sentiment feature
    module_8_deploy/                          ← M8: FastAPI app + Dockerfile
  module-3/class_X/submissions/<Name>/...    ← (existing structure)
  ...

```

Students fork or clone this. By M8 their fork is a deployable repo.

8. What each student type sees

8.1 Type 2 (A2 English, first-time laptop)

- Bronze assignment is a Colab notebook with code already written, 5 specific TODOs to fill in.
- Each TODO has 2-3 sentence inline explanation in plain English.
- Worked example linked to the equivalent YouTube tutorial, timestamped to the relevant 3-minute clip (not the full 30-minute video).
- “Why this matters” 60-second video at top of every assignment, recorded in Russian/Uzbek by the mentor (one-time recording, reusable).

8.2 Type 1 (comfortable, wants stretch)

- Gold assignment is a problem statement and the dataset.
 - Encouraged to skip the lecture videos and self-study from docs.
 - Stretch: contribute one extension (e.g. *try XGBoost, add a calibration plot, propose an alternative target*) and write 200 words on the result.
 - Public credit on the website’s “Contributors” page if their extension is merged into the cohort’s pipeline.
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9. Risks & how we mitigate

Risk	Mitigation
One dataset gets boring	We don’t change the dataset — we change the lens . M5 looks at customers, M7 looks at reviews, M8 looks at deployment. Same data, different angles.
Olist is non-Uzbek	Add 1-slide context per module connecting Olist insight to a local equivalent (e.g. “if Uzum had this data...”).
Some students prefer their own dataset for capstone	M8 capstone has an opt-out: students who want their own dataset can use one, but must demonstrate the same 6-stage pipeline.
Brazilian Portuguese in reviews	M7 explicitly uses a multilingual model. Translation isn’t required — the embedding handles it.
Olist data quality changes (Kaggle removes the dataset)	Mirror the data once into the cohort repo’s <code>olist_data/</code> folder. We’re already doing this — see §7.

10. Decision log

- **2026-05-04** — locked Olist as the compound dataset. Reason: dirty, multi-shape (numeric/cat/text/dates), has photos for M6/M7, multi-table for M2 merge skills, recognised name on CVs.
 - **2026-05-04** — primary headline target = `is_late`. Reason: imbalanced, business-relevant, improves with every module.
 - **2026-05-04** — assignments two-tiered (Bronze / Gold) starting M3. Reason: solves the Type 1 / Type 2 split without splitting cohorts.
 - **TBD** — mid-term covers M1-M4 only. Final covers everything. (Confirm with program partner.)
 - **TBD** — final exam includes a live demo of the deployed M8 system. (Pending faculty buy-in.)
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This document is the contract. Any deviation in module rewrites should be flagged here.